



Review Article

Blockchain-based governance models supporting corruption-transparency:
A systematic literature reviewMohammad Mustafa Ibrahimy^{a,*}, Alex Norta^{b,c}, Peeter Normak^a^a School of Digital Technologies, Tallinn University, Narva mnt 25, Tallinn 10120, Estonia^b Baltic Film, Media and Arts School, Tallinn University, Tallinn, Estonia^c Dymaxion OÜ, Tallinn, Estonia

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ABSTRACT

Corruption and lack of transparency remain critical challenges in governance systems around the world. These issues are often perpetuated by centralized systems and their manipulation by system administrators. Furthermore, the lack of data ownership and the monetization of user data by tech companies further increase concerns about transparency. In light of these concerns, this study aims to review existing blockchain-based governance models and identify best-practice governance models focusing on corruption transparency, their characteristics, and components. The research will also examine the role of a token economy in addressing trusted third-party issues related to asset ownership management. Furthermore, we discuss the effect of smart contracts, blockchain, decentralized autonomous organizations (DAOs), Web 3.0, and multifactor challenge set self-sovereign identity authentication (MFSSIA) as modern technologies to combat corruption and achieve transparency in the public sector. To achieve this, we conducted a systematic literature review (SLR) comprising peer-reviewed journals, proceedings, and book chapters published between 2012 and 2023. Using the SLR methodology, 45 primary and supporting studies have been selected for result extraction and analysis. Finally, we discovered seven blockchain-based governance models with their characteristics and primary components.

1. Introduction

Blockchain is a decentralized and distributed database of cryptographically linked records, known as blocks, arranged in chronological order and shared between computer network nodes for digital information storage [1]. Smart contracts, as a computer program [2], are stored on a blockchain that runs automatically without involving any third-party intermediaries [3]. By eliminating third-party intermediaries, smart contracts play a pioneering role in automated business participation by making collaboration processes more transparent and efficient in terms of time and money [4]. Web 3.0, the third generation of the Internet, is a user-centered web that aims to safely decentralize user data, including money, ownership, and identities, through the use of blockchain technology, providing complete control for users instead of centralized entities to manage their data [5]. Blockchain is considered the driving force of Web 3.0, providing data ownership to users and allowing them to store and manage their data in a decentralized way through self-sovereign identity (SSI) [6]. Mean-

while, multifactor challenge set SSI authentication (MFSSIA) is a critical aspect of building trust between entities such as systems, devices, organizations, and humans in the emerging machine-to-everything (M2X) economy. It offers self-sovereign, flexible, and configurable multifactor challenge set-based identity authentication for the secure participation of humans and machines in electronic market transactions [7].

SSI is a user-centric, user-controlled approach, a new identity ecosystem, and a digital identity service in which users, or entities to whom the identity concerns, have control over their data and manage their digital identities without relying on a central organization, or entity. SSI is used in the virtual or physical world [8,9]. To correctly identify a user and provide services, a secure and reliable identity is required [10]. Numerous SSIs rely on public cryptography. Each user owns a unique private key (symmetric key) that should be kept secret and is paired with a public key (asymmetric key) [8].

Several studies [11,12] suggest that an SSI solution does not inherently require the use of blockchain technology. However, this technol-

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ogy provides a robust foundation due to its numerous technical benefits and serves merely as a facilitator in SSI implementations. However, it should not be considered as the ultimate solution for SSI. Blockchain technology, by forming the technical basis for the SSI concept, enables whistleblowers to report wrongdoing, suspicious activities, fraud, and corrupt practices anonymously without disclosing their identity to investigators [13]. Therefore, blockchain has the ability to empower individual users by asserting their self-sovereign identities, which is the strength of this technology in boosting governance mechanisms and reducing corruption [14].

The study in Ref. [15] compares SSI systems, categorizing them into blockchain-based, such as Sovrin and uPort, and nonblockchain-based, such as the project 'I reveal my attributes' (IRMA). While blockchain-based SSIs offer robustness against collusion, they pose architectural challenges. Non-blockchain SSIs are easier to deploy but risk schema manipulation if managed by a single entity.

Token economies are a new form of decentralized digital economy powered by blockchain that incentivizes participation in blockchain networks and platforms. This reduces the gaps between the needs of companies and participants with respect to data utilization, with high security and privacy on blockchain data sharing platforms. Furthermore, token economies empower individuals by giving them digital control over their identities, credentials, work products, and assets [16–18]. On the other hand, blockchain technology facilitates digital public services and social innovation tools through tokenization and digital asset transactions that improve socioeconomic interactions, citizen participation, public governance, and democratic processes [19]. Therefore, token economies are a key mechanism for incentivizing participation and desired behaviors within decentralized blockchain networks. As such, designing effective token economic models is an important governance consideration.

Corruption occurs in both public and private sectors. Therefore, individuals and private corporations are inclined to pay bribes in return for services obtained [20]. The fight against large-scale corruption is difficult with information and communication technology (ICT) systems, since corrupt elites design and control the system in their favor. In contrast, emerging new technologies, such as blockchain, promise to be tamperproof, transparent, traceable, and auditable [21]. Moreover, the use of smart contracts facilitates the automation of processes in the public sector without involving any third party, and therefore, the information is safe from manipulation [22]. Therefore, the adoption of smart-contract blockchain technology is considered one of the major tools for developing countries to reduce corruption, improve transparency, and eliminate central authorities and intermediaries in systems associated with human interference [21,22].

Distributed ledger technologies (DLTs), especially blockchain technology, hold significant potential for public institutions to revolutionize avenues for citizen participation and revitalize democratic processes [23]. This potential is particularly evident in the context of a digitally enhanced democracy, which can leverage DLTs to inherently uphold principles such as transparency, trust, and autonomy. More specifically, DLT can serve as a catalyst for participatory digital democracy, providing innovative governance mechanisms such as reliable data storage and reliable computations [24]. Therefore, in the context of e-democracy, DLT provides secure, transparent, and efficient methods for recording and verifying transactions and interactions. This enhances various aspects of e-democracy, such as online voting, public consultations, and digital identities.

Several studies propose various blockchain-based architectures and models to solve tracking problems and contract management [25], ensure the quality of logistic services [26], accelerate the procurement process, and reduce transaction costs [27]. These models also aim to facilitate better management and increase citizen participation [23,28], ensure trust among stakeholders [29], increase the transparency of service processes in all government transactions [30,31], and improve performance by removing intermediaries and making payment pro-

cessing more efficient through smart contracts [32]. Furthermore, they allocate resources automatically in the management of public-funded projects [33], fight corruption [34], and prevent public funds embezzlement as the biggest corruption [21].

The existing literature focuses mainly on different proposed blockchain-based architectures/models that are used for various purposes. However, none of these studies identify the best-practice governance model that must be adopted in the public sector to improve transparency and reduce corruption.

The state of the art in Ref. [35] shows that different technological means and e-democracy models increase e-participation in governance processes, thus increasing the transparency and accountability in the organization. Similarly, several e-democracy models by Mørøe et al. [35] have been discussed on citizen participation and improving e-democracy, while there is a gap in tackling corruption, achieving transparency, increasing citizen participation, and empowering them using modern technologies such as blockchain, decentralized autonomous organizations (DAOs), Web 3.0, and MFSSIA [7].

This study aims to describe existing blockchain-based models, their characteristics, and their components. Furthermore, this study conceptualizes the application of modern technologies mentioned above and their contribution to the public sector. We further describe the role of the token economy and its application in the public sector. To achieve this, a systematic literature review (SLR) is performed based on Kitchenham and Charters's guidelines [36] in software engineering. This study answers the following key questions:

1. What are the existing blockchain-based governance models?
2. What are the main components and characteristics of these governance models?
3. What is a blockchain-based token economy and its role in eliminating trusted third parties in transferring ownership?
4. To what extent does the use of smart contracts, blockchain technologies, DAOs, Web 3.0, and MFSSIA affect the public sector to achieve transparency?

The rest of this study is organized as follows. In Section 2, a comprehensive overview of the relevant literature with the latest findings is presented. Section 3 provides the SLR methodology and Section 4 summarizes the research findings followed by a discussion in Section 5. Finally, Section 6 presents the conclusions, limitations, open issues, and future work.

2. Background and related work

To elaborate blockchain-based governance models, we first describe the concepts of blockchain and governance in Section 2.1. We give an overview of the blockchain application in the public sector with its associated challenges in Section 2.2, and blockchain-based models in Section 2.3. In Section 2.4, we discuss tokens, tokenization, and token economy followed by a description of smart contracts, blockchain technologies, DAOs, Web 3.0, and MFSSIA in Section 2.5.

2.1. Blockchain & governance concepts

Blockchain (also known as the distributed ledger) is a distributed database, decentralized peer-to-peer (P2P) network that provides transparency, immutability, authenticity, and security [37–39]. A distributed ledger means a ledger that is shared between all peers/blocks on the network, where each peer has a copy of the complete ledger on that blockchain [40]. This technology is secure, decentralized, and distributed over the Internet, removing intermediaries such as banks and other trusted third parties [41–43].

In the blockchain, a block is a collection of different data where these data are added to a single block, connecting all blocks chronologically in a continuing series to ensure the authenticity and immutability

of data. Each block must contain three specific elements: (1) the block header, (2) a set of transactions, and (3) information about the validity of the block transactions. All transactions are secured by a public cryptographic key, which is validated by all peers participating in the network. Once the data has been added to the blockchain, they become unalterable and immutable. The first block on the blockchain is termed the Genesis Block [40,44].

Governance is defined as processes that are governed by the laws, norms, or powers of a government, business, or network. Blockchain governance models describe 'the rules by which assets managed in the ledger are created and how the ledger is maintained' [45]. Governance provides clearly defined roles and duties and ensures that agreed processes reach decisions [46]. Governance could be approached from two perspectives: (1) at the firm level, a system of rules and structures could be applied to the company's managerial decisions, and (2) at the government level where rules and structures are imposed on businesses, affecting the scope and the way these managerial decisions are made [47]. Blockchain governance decisions are analyzed at the micro, meso, and macro levels such that governance decisions at one level do not stand alone; rather, they are interconnected levels of blockchain governance [48].

2.2. Blockchain application in the public sector and its relevant challenges

The government power is abused by exercising corruption at all levels from state to local, and this corruption ranges from small-scale bribes to large-scale bribes [43]. However, governments around the world are working hard to promote social empowerment [34]. To this end, governments and other public organizations are exploring new ways of applying blockchain technologies around the world to achieve smart governance, minimize cost, and improve work efficiency.

Blockchain offers lower infrastructure costs under certain circumstances, primarily by eliminating the need for intermediaries and facilitating P2P transactions. However, there are also significant costs associated with blockchain technology, particularly in terms of energy consumption and the costs of purchasing hardware and software systems to operate a decentralized network. The energy demands of blockchain technology vary significantly depending on the specific design choices made. Any increase in the energy consumption of a particular blockchain must be considered in relation to the cost savings it offers [49,50]. The energy demands of blockchains are closely related to the type of blockchain and its consensus mechanism. Public blockchains, for instance, are characterized by high energy consumption, while private or federated/consortium blockchains typically have lower energy demands. Among common consensus algorithms, Proof-of-Work (PoW) is associated with high energy consumption, while Proof-of-Stake (PoS) is associated with lower energy use [51]. Blockchain technology advances democratic innovation and promotes citizen participation in the public sector through streamlining processes and securing recorded data, thus guaranteeing public transparency [28]. Governments must increase transparency levels to gain the trust of citizens and achieve good governance. Therefore, the absence of good governance leads to corruption, creating opportunities for corrupt practices [43].

There are several barriers and challenges that prevent the widespread adoption of blockchain technologies. These barriers are the technological limitations of blockchain and smart contracts, organizational complexity and misalignment caused by blockchain implementation, lack of system integration, interoperability issues, lack of a governance model, as well as cybersecurity risks that will impact the adoption of blockchain technologies in organizations [41]. Moreover, Batubara et al. [52] classified the main challenges of blockchain adoption into three categories: (1) technological, (2) organizational, and (3) environmental. The technological aspects include security, scalability, and flexibility. The organizational challenges are acceptability issues and the need for

a new governance model, and, finally, the environmental barriers are a lack of legal and regularity support.

The necessity for widespread adoption of blockchain technologies depends on the specific goals that need to be achieved or realized. While other technologies may suffice in some cases, blockchains offer unique advantages that are unattainable with other technologies. For instance, if the objective is merely to automate a process, a trusted third party might render the use of blockchain unnecessary. However, for goals such as creating a system that is transparent, tamper-proof, and decentralized, blockchain could potentially be the optimal solution. In addition to these benefits, the elimination of trustworthy third parties could also lead to a reduction in costs and corruption. This is particularly relevant in financial applications where parties are generally risk averse and prefer not to rely heavily on trust assumptions, making blockchain technology well-suited in general [53].

2.3. Blockchain-based governance models

Blockchain-based governance models can be classified in Table 1 based on several common characteristics, including the type of blockchain (open or authorized), the blockchain platform used, the specific challenges addressed, and their application in various domains. They are context-specific and have been implemented in diverse sectors such as supply chain management, public procurement, real estate, and government services. This diversity in application underscores the adaptability and versatility of blockchain technology in addressing unique challenges across different sectors.

Mohite and Acharya [30] highlighted the use of Hyperledger Composer, a rapid prototyping tool for blockchain applications. By defining a business model and deploying it on the Hyperledger fabric, this tool facilitates the creation and testing of a blockchain-based system that enhances transparency in government transactions and fund tracking to find out how the budget is being spent.

Tasca and Piselli [60] compared the two alternative types of governance models: one is the so-called informal governance models (for example, Bitcoin, Ethereum), and the second is known as chain-based models (for example, EOS, NEO) that emerge from more complex governance systems. The first type offers the participation of a core group of developers who formulate the proposal for protocol amendments and to whom the duty of acquiring the classified consensus of stakeholders is appointed. The latter type of governance for two reasons differs from the initial one. First, the voting processes are open to all coin holders, making the system more democratic. Second, the voting result is directly included and executed by the system.

De Rossi et al. [61] dimensionally classifies blockchain governance models into four types. (1) public permissionless: Anyone can read, write, and commit; (2) private permissionless: Anyone can read and commit, but only authorized participants have the write privilege; (3) public permissioned: Read permission is restricted, but write privilege is open to anyone and only authorized participant commits and (4) private permissioned: Read permission is restricted, and write permission and commit are open only to the network. Therefore, the first two are considered open by their ownership, whereas the latter two are considered closed.

Vinet and Zhedanov [62] argued that the greatest advantage of each governance model is also its greatest disadvantage. Dursun and Üstündağ [57] state that blockchain-based governance models according to their execution are classified into off-chain and on-chain governance. Off-chain governance is similar to classic social governing structures such as direct democracy, which includes core developers, miners, users, and organizations in a process that is mainly based on human interactions for consensus on change proposals, and then the blockchain software is manually updated. Off-chain models are somewhat centralized due to dominant actors; thus restricting common users who do not have sufficient knowledge on technical or financial issues to influence network decisions. The off-chain model is time-consuming

Table 1

Typology of governance models based on their common characteristics.

No.	Source	Category	Platform	Aspects Being Enhanced	Application Context
1.	[21]	Semi-decentralized	Hyperledger Fabric	• Transparency & Auditability • Preventing public funds embezzlement	Government sector
2.	[25]	Decentralized	Ethereum	• Transparency • Management of contract	Public Procurement
3.	[28]	Centralized	Neural blockchain/RETIS	• Citizen participation • Rewarding citizen participation via token	Public affairs
4.	[31]	Decentralized	Ethereum	• Transparency	Supply Chain
5.	[32]	Decentralized	Ethereum	• Traceability • Transparency, and trustworthiness	Supply Chain
6.	[37]	Centralized	Hyperledger Fabric	• Document verification • Efficiency of issuing process	Government ID verification
7.	[41]	Quasi-centralized	Hyperledger Fabric	• Transparency of data, and trust • Protection of sensitive data • Traceability, and efficiency	Supply Chain
8.	[44]	Centralized	Neural blockchain/RETIS	• Openness and transparency • Fighting against corruption	Public procurement
9.	[54]	Decentralized	Ethereum	• Security level • Simplifying administrative procedures	Government administration
10.	[55]	Centralized	Hyperledger Fabric	• Speed and efficiency of the rental process • Security and fraud protection • Process automation	Real estate
11.	[56]	Centralized	Hyperledger Fabric	• Auditing • Eliminating centralized entities in system	Auditing
12.	[57]	Decentralized /Centralized	Ethereum/Hyperledger BESU	• Participation in the governance process	Decision-making
13.	[58]	Centralized	Hyperledger Fabric	• Verification of academic credential • Transparency	Education sector
14.	[59]	Decentralized	N/A	• Citizen participation • Transparency, and trust	e-voting

and runs slowly because it is conducted outside the blockchain with several individual actions and is conservative with gradually improving. The rules in off-chain governance are easily adjusted to changing circumstances, and these rules are unclear and malleable, but only the involvement of a third party authority enforces them individually. Conversely, on-chain governance is a recent concept; the changes take place without the requirement for a hard fork. The consensus is often reached by voting through the protocol. The results of the voting are controlled algorithmically, and their automated execution is embedded directly into the protocol. Vinet and Zhedanov [62] emphasized that on-chain governance has more formal rules, it is strict, predictable and most efficient than off-chain governance because on-chain governance is explicitly codified and enforced automatically by the underlying technology according to defined processes. On-chain rules are not easy to adjust to a changing environment.

The governance model is adopted by blockchain technology in IT systems to further develop traditional systems into a multi-centered or decentralized system; the system will operate below the governance model based on the degree of trust among participants [63,64]. Blockchains are based on a consensus of various types, each of which has its own technical features and governance model [65]. Furthermore, blockchain provides a new technological means for designing a governance model based on computerized trust management through smart contracts [66].

Baset et al. [46] emphasized that business models and governance structures are dependent on each other to properly control the operations of blockchain networks. A well-planned governance model creates balance among involved institutions, who could be considered competitors, co-creators, or collaborators at various times. IT solutions need governance support, and the governance process needs IT solutions.

2.4. Token, tokenization, and token economy

Blockchain use cases extend beyond managing cryptocurrencies. In particular, the token economy, where tokenized assets are exchanged between different blockchain ledgers, is gaining popularity [67]. The concept of using digital tokens for asset ownership is known as the token economy [68]. A token economy is defined as a method of encouraging desired behavior [69], a complex reinforcement system that provides participants with a medium of exchange (i.e., tokens) to trade different goods, services, or privileges [70]. Moreover, the token economy is an ecosystem (reward system) composed of three key elements: tokens, desirable behavior, and rewards. In this ecosystem, tangible or intangible values circulate between the desired behavior and the token, which serves as a reward for network participants [71].

2.4.1. Token

Token and token economy (crypto economy) are closely related to smart contracts [72]. A token, in the token economy, is a unit of value that is coded and run on any blockchain [73]. Mougayar defined a token as “a unit of value that an organization creates to self-govern its business model and empower its users to interact with its products while facilitating the distribution and sharing of rewards and benefits to all of its stakeholders” [74]. Crypto tokens are classified in two ways: (a) by their usage, which is fungible or non-fungible tokens (NFT), and (b) by their type of implementation, which includes payment (currency), utility, security [75], governance [76], platform, transactional [77], NFTs and soul-bound tokens (SBT) [78].

1. **Payment tokens** (also called cryptocurrency or consensus): They serve as a store of value and a unit of measurement, and digital

assets in the core of blockchain platforms such as BTC (Bitcoin) or ETH (Ethereum) [72,79]. BTC and ETH are the two most widely used currency tokens that are spent and exchanged as traditional fiat currency [77].

2. **Utility tokens** (also known as applications or user coins): They refer to tokens that represent a type of tokenized rights or access given to a service or a product on a blockchain network [77] such as a gift card. An example is StorjCoin, which provides access to a distributed storage system.
3. **Security tokens** (also known as equity tokens): These types of token represent ownership shares of a company or DAO platform, or digital representations of traditional securities such as equities, bonds and options that give the token holder the rights to future profits of the company (e.g., tZERO) [72,79].
4. **Governance tokens**: Governance tokens are a type of cryptocurrency that represents voting authority on a blockchain project and have recently been mainly incorporated into decentralized finance (DeFi) projects as a way of allocating rights to users to ensure decentralization. Furthermore, governance tokens have economic value that is traded on both centralized and decentralized exchanges (CEX and DEX) [76]. On the other hand, governance tokens represent a stake in a DAO, allowing for the distribution of control among stakeholders [80].
5. **Platform tokens**: They allow the operation of decentralized application stacks and a diverse set of mechanisms and use cases. These tokens are usually used in smart contracts or underlying blockchain networks to offer various functionalities throughout the blockchain infrastructure. A key feature of platform tokens is the provision of rewards to platform users and, in some cases, the reimbursement of transaction fees. An example of a platform token that is designated as a stable coin is MakeDAO's DAI token.
6. **Transactional tokens**: These tokens act as units of account and generally have characteristics similar to traditional currency. Transactions using transactional tokens do not require intermediaries or central authorities such as banks or clearinghouses. They are also used to facilitate transactions such as currency exchanges [77].
7. **NFTs**: These tokens are issued by the issuers (or artists) as proof of ownership to the buyers and are transferable. These tokens are stored in the issuer's account as proof of ownership. When a buyer purchases the artifact, the token is transferred to the buyer's account, which now proves its ownership. As the new owner of the artifact, the buyer can transfer the token to others as proof of their ownership.
8. **SBTs**: They are nontransferable tokens that have characteristics similar to NFTs, with the distinction that they cannot be transferred to other accounts once they have been issued to a particular account, and could be utilized in conjunction with NFTs to verify ownership of an artifact [78].

Tokens represent ownership rights, stocks, certificates [81], the possession of a house, the identity of an individual, the right to consume a service, organizational decision-making [82], existing digital or physical assets, and access rights to assets and permissions in the digital or physical world [83]. In addition, tokens serve as a means of payment where the form of payment is digital, physical, virtual, or legal [84], and provide additional functionality against traditional funding mechanisms [85], thus playing significant roles in granting access to a service, the right to contribute to a community, and regulating governance through voting rights [86].

2.4.2. Tokenization

Tokenization refers to the process of converting the value of a physical (e.g., land, building, or inventory) or non-physical asset (e.g., trademarks, copyrights, or licenses) into a digital form (that is, the token) that is produced by an algorithm, represented digitally and exchanged

through a blockchain network [87]. In the blockchain context, tokenization signifies digitally representing physical or electronic assets on the blockchain [88]. The notion of tokenization demonstrates that by eliminating intermediaries, such as notaries and lawyers, transactions become more efficient in terms of time and cost savings [87]. Blockchain-based tokenization has several advantages, such as democratizing the investment market by allowing fractional investment with lower costs, and execution of transactions on a blockchain without involving any third parties that offer cheaper and faster transactions. It also improves market efficiency by eliminating the time and calendar restrictions of the global market and unlocking the value of previously illiquid assets (e.g., houses, cars, or land) and allowing them to be traded immediately and cheaply [89]. Moreover, other advantages of tokenization include cheaper and faster transaction processing, decentralization, security, flexibility, and transparency. In addition to these advantages, tokenization also has some disadvantages, including regulatory and legality issues, and technical barriers caused by using DApps [88]. The tokenization process of physical assets imports a positive change, in which the assets are split into smaller parts to be owned, traded and exchanged without the need for any intermediary [90]. Thus, the costs of intermediaries are eliminated through tokenization [91].

2.4.3. Token economy

Previous studies on token economy came mainly from the fields of psychology, pedagogy [83,92,93], psychiatry, clinical psychology, education [68], and treatment [16]. In the fields of psychology and pedagogy, the token economy is defined as 'a management system that uses tokens as an incentive to promote the positive behaviors of the target while devaluing the negative ones of the target' [94].

In the fields of psychiatry, clinical psychology, education, and mental health, the token economy has been used for many decades, indicating that the application of operant conditioning (i.e., a method of learning in which behavior is modified through the use of rewards and punishments) offers promise to achieve different treatment and educational goals [95]. After the introduction of bitcoin and blockchain, and their importance in crypto-economics, the studies on the token economy gained new momentum and the token economy received much attention for its financial importance [93,92]. However, a general definition of token economics in the blockchain space is still missing. Lesche et al. define a token economy in the context of blockchain and DLT research as an 'incentive system based on cryptocurrencies with the goal of achieving the desired behaviors using blockchain/DLT' [68]. The sudden growth of the cryptocurrency market and the adoption of Bitcoin by some governments (e.g., El Salvador) allows token economics (also known as tokenomics, or cryptoeconomics) to be used for organizational information sharing without requiring third-party entities such as banks or payment systems [88]. The economics of tokens defines the utility of each token and how the token is used to encourage good behavior in a blockchain network [68].

A token economy offers several advantages, including superior liquidity, faster and less expensive transactions, greater transparency, and greater accessibility [67]. The blockchain-based token economy uses tokens to achieve a real mapping of real economic activity to digital economic activity and uses smart contracts to automatically distribute benefits and timely implementation of laws, regulations, norms, and commitments [17]. Existing studies on blockchain-based token economics focus mainly on three aspects. First, the design of token mechanisms focuses on how to design a valuable token. Second, the design of particular economic scenarios. Third, the use of tokens in the actual business schema, for example, the joint utilization of ICO and tokens [70].

In DLT-based token economy instances, tokens are created and managed either via a DLT protocol or via smart contracts. Tokens, that is, native tokens, are created via the DLT protocol and are often utilized in DLT systems to reward participants for contributing computational resources to the operation of a DLT system. Actors implement custom business logic using smart contracts to create and manage tokens that

represent any assets on the DLT protocols [68]. Multiple instances of the token economy are created from the microeconomic perspective. There are two main options for creating an instance of a DLT-based token economy: first, using custom tokens of a distributed ledger and second, creating tokens on an existing distributed ledger [91]. Token-economy instances are either centralized or decentralized. In centralized token-economy instances, usually interoperability is enabled by central actors among themselves, so that tokens are transferred among them over the digital platforms. In a decentralized token economy, multiple actors participate in the decision-making for important tasks in asset ownership management without relying on a central authority or controlling body. Decentralization of token-economy instances reduces reliance on central actors and their digital platforms by addressing drawbacks related to the central actors' involvement in asset ownership management and allows direct transactions between actors without involving any intermediaries such as banks, thereby improving cost efficiency and preventing performance bottlenecks. Therefore, before instantiating a decentralized token economy, the following drawbacks must be addressed [68]:

1. Double-spending of tokens must be prevented.
2. Tamper-proof storage of account balances and transactions.
3. Coordination of parties to reach consensus.

Transferring asset ownership between individuals or organizations is performed by a trusted third party such as a bank, or a notary, to increase the reliability of the transfer process, e.g., traditional currency, company shares, or rights of use. The involvement of such a party gives rise to several disadvantages, such as high costs, slower processing times, and the existence of a single point of failure (SPOF). The presence of such drawbacks leads to decentralization and automation of services provided by a third-party authority. Emerging advanced technologies enable agents to represent and manage ownership of assets digitally using tokens in a fraud-resistant way on a decentralized digital platform without the need for intermediaries. This minimizes drawbacks such as SPOF pertaining to trusted third parties and thus enables the token economy. The token economy has the potential to revolutionize the way we exchange value and transfer ownership of assets, providing greater security, efficiency, and flexibility than traditional methods. Decentralized digital platforms should prevent double spending problems and ensure that users cannot use the same tokens multiple times simultaneously [91]. DLTs are used to decentralize digital platforms to manage asset ownership in token economy instances [68]. Similarly, the existing blockchain token-economy paradigm consists of several drawbacks such as the lack of incentive mechanisms, business applications, and the value of virtual tokens. These shortcomings minimize the willingness of consumers and merchants to make a profit [70].

Fujimoto et al. [67] proposed a new approach called ConnectionChain for interworking blockchains that offers a better infrastructure in the token economy settings to trade tokenized digital assets. The goal of ConnectionChain is to provide solutions for the following problems:

1. The diversity of choices on blockchain implementations is challenging to manipulate these ledgers programmatically due to incompatibility between communication protocols and transaction data formats.
2. A smart contract is not used as a trusted third party for trade that includes transactions on various blockchains because a smart contract only controls assets on the same blockchain. To ensure the secure issuance and trading of tokens, legal regulators are encouraging financial institutions to comply with the 'security token' requirements.

A token economy is an economic system that uses cryptography instead of a trusted third party [85], a token reinforcement system defined by tokens in which tokens play the role of medium of exchange instead

of legal currency, and act as a bridge between the real economy and the fictitious economy. By using the DAO community, which includes developers, investors, users, and other stakeholders who are interested in creating and participating in DAOs, or DAOs themselves as a carrier for the token economy, it is possible to incentivize active coordination between companies. This is achieved by offering valuable rewards to participants, which help improve information exchange, data processing, employee motivation, and collaboration between different companies and departments [16]. In the token economy, there is no need to share the original transaction data. Instead, the data demander simply inserts the smart contract into the data provider's original transaction data set to analyze and then obtains the analysis results. The token economy involves features of consensus, such as value transfer and smart contract execution, and economic activity that has a rapid information transmission mechanism, strict execution rules, and efficient execution. These features improve government operations, streamline the government approval process, and reduce taxes, fees, and administrative powers that significantly reduce the pressure on businesses and restore their vitality [17].

A well-designed token economy also takes into account the number of tokens issued, the compensation of participants, the redistribution of wealth, and the continued participation of users in services. In a token economy where a variety of crypto-tokens are actively traded, financial institutions must view crypto-tokens as complementary to existing currencies and provide enough liquidity to the market [96]. A token economy is considered an effective funding mechanism that ultimately enables decentralized governance of technology platforms. A token economy should offer multidimensional token functionality, such as payments between participants, sustainable stores of value, and foundation financing [85]. Steemit is a token economy-based online content publishing system in which a unique type of token economy is used to reward content creators and promoters and punish irresponsible or malicious behaviors [92]. Steemit is one of the first and most successful platforms in the blockchain-based online social network ecosystem that calculates the value of contributions through three native tokens, each with a specific purpose: (1) STEEM (block letters to avoid confusion with the Steem blockchain), (2) Steem Blockchain Dollar (SBD), and (3) Steem Power (SP).

1. **The STEEM token:** It is the liquid cryptocurrency at the base of the token system that Steem users transfer to each other as a form of payment. STEEM is traded with other cryptocurrencies on different exchanges or more traditional currencies such as US dollars, or exchanged by users as a form of payment. STEEM token is the fundamental unit of account on the Steem blockchain from which the other tokens derive their value.
2. **SBD:** It is designed to provide a stable value cryptocurrency. SBDs are created by a mechanism similar to convertible notes, which are short-term debt instruments that are converted to ownership at a rate determined in the future. Steem rewards contributions with cryptocurrency, enabling creators to monetize their content. The Steem blockchain has a Delegated Proof of Stake (DPoS) consensus mechanism, where users with a high stake have more power over the distribution of the tokens than users with a smaller stake. STEEM is viewed as ownership in the Steem community, whereas SBD is viewed as a debt denominated in any commodity or currency. As in STEEM, SBD is bought and traded outside the Steemit platform through exchanges.
3. **SP:** It is an access token that grants its holders voting power within Steem. Users convert their STEEM to SP by committing them to a vesting schedule, a process called powering up. SP is nontransferable and is withdrawn by converting the SP back to liquid STEEM in 13 equal weekly payments, a process called powering down. SP is the equivalent of the market shares of Steem. As in common shares, if the value of the company increases, so does the value of the users' shares. SP cannot be acquired or traded, a substan-

tial difference from the other currencies. In short, the only way to get SP is either as a reward or by converting other tokens into SP. While users are not required to spend the STEEM or SBD for participation in daily activities (e.g. posting content), they need to hold some SP because the amount of SP determines the number of social actions (posts, comments, and votes) a user performs. Moreover, SP influences activity in the network in different ways. For example, SP plays a key role in the reward process, posts voted on by users owning a large volume of SP gain more visibility, and the top posts also get a larger share of the reward pool. If a user is not interested in social activities, the STEEM and SBD tokens are exchanged for services and/or goods with other users or on trading platforms [97,98].

Furthermore, Aistov et al. [85] designed a token economy for the Open Charging Network (OCN) of the Share&Charge foundation to provide enough financing, support distributed governance of the organization, and incentivize early participation. The proposed token economy should meet several business goals and stakeholder requirements, as well as follow some general design guidelines for token economies. Thus, the token economy model for the OCN has to:

1. Provide finance for the platform and its operators.
2. Offer profitable long-term investment opportunities.
3. Provide value to early participants.
4. Support a decentralized governance model.

As the current centralized design of the digital economy created powerful oligopolies in major online service markets, therefore, a token economy that is built on decentralized business logic and data layers poses itself as a credible alternative. Intermediaries as trusted third parties build business models to monetize their roles either directly or indirectly through fees or monetization of collected data, and it is unlikely that their roles will be obsolete with tokenizations in enabling consumers to participate. Therefore, to successfully create a token economy, technological, standardization, and governance aspects must be taken into account [91]. In terms of business challenges and the intersection of business and law, the biggest problems companies face in the token economy is a lack of infrastructure in the form of exchanges, resulting in low liquidity and thus a volatile market, and a lack of clear regulations that significantly impede the further development of business. The first problem will be solved as soon as the main players of the crypto industry and the traditional industry work together to develop the required infrastructure [68].

Token economies are implemented in virtual projects and use cryptocurrency-based incentive systems that reinforce and generate desired behavior in the blockchain ecosystem. These projects are typically public blockchain platforms such as Ethereum, Aeternity, and Tezos, and attract funding through an ICO or an initial exchange offering [99]. Lympo¹ is a blockchain-based ecosystem designed to allow users to monitor their health and sports data while incentivizing them to adopt a healthier lifestyle by offering LYM tokens as rewards. Lympo was one of the early applications affiliated with the Samsung blockchain wallet, and users withdraw LYM tokens directly from their Samsung blockchain wallet. The application includes daily running and walking challenges that users must complete to receive LYM token rewards, which are used to purchase quality sporting goods from an online store called Lympo Shop. A token economy such as Lympo has a strong incentive because it allows users to own and monetize their data by providing their personal data to an industry where these data are used to develop systems that promote physical fitness [100–102]. The token economies are divided into two dimensions: the design of a reward distribution structure for nodes and the design of the currency issuance mechanism. Designers of

a token economy search for ways to prevent undesirable behavior that devalues the ecosystem in terms of cost and revenue. In a token economy, the high or low reward does not directly affect the value of the entire token ecosystem [71].

2.5. Blockchain technologies, smart-contracts, DAOs, Web 3.0, and MFSSIA

Blockchain technologies, along with their subset technologies such as smart contracts, DAOs, and Web 3.0, are the pillars of a new digital era. These technologies are paving the way for a future where transactions are secure, transparent, and decentralized, eliminating the need for intermediaries and improving user control. The concept of MFSSIA further strengthens this vision by providing a secure framework for the participation of individuals, organizations, and systems in e-Commerce transactions. As we explore these technologies more thoroughly, we will explore each of them one by one as follows:

2.5.1. Blockchain technologies and smart contracts

Blockchain technology is a distributed database that keeps a record of all transactions in a network. One of the key aspects of blockchain technology is its ability to facilitate communication between untrusted parties without requiring the presence of a middleman or trusted third party [103]. In traditional systems, transactions occur centralized and require the participation of a trusted third party. This approach poses security concerns since it creates a SPOF, as well as high transaction fees [104]. The adoption of blockchain and smart contracts will minimize fraud and errors in all recorded information on the network, enabling better public service delivery with greater efficiency, transparency, and fairness [105]. Blockchain, along with other decentralized digital technologies, eliminates bureaucracy and paperwork by eliminating physical interactions with government offices. Governments are experimenting with three main blockchain functionalities: notarization, shared databases, and workflow automation. Blockchain-based notarization services are more developed, while more disruptive solutions face implementation challenges due to current administrative processes and regulatory non-compliance [106].

The transparency provided by blockchain transactions is instrumental in detecting and, ultimately, reducing or eliminating corrupt activities. Blockchain is designed to address three main issues. Lack of transparency: The current system is generally not transparent because it is not easily observable to the public or to other parties involved. Slow process: The process of obtaining a legally binding purchasing contract takes 3 to 6 months after the title has been approved. Complexity leading to inefficiencies: Each participant involved in the registration procedure, including sellers, buyers, banks, and real estate agents, aims to ensure that the process functions properly, as the transaction value is significant. This results in the overly intricate and ineffective registration process [90]. Blockchain technology is characterized by several key features, including decentralization, transparency, robustness, auditability, and security [107].

Blockchain-based applications operate autonomously; they are highly effective in facilitating and simplifying transactions between people, companies, governments, and even machines [108]. Hyvärinen et al. [109] developed and evaluated a blockchain-based prototype system, specifically tailored for the Danish tax authority's context, with the goal of reducing tax fraud and enhancing transparency in dividend flow. Their system incorporates crucial features of blockchain-based solutions, including decentralization, transparency, immutability, and automation. However, it is not a completely trust-free system, as it relies on the cooperation of banks and the trust of institutions that issue tokens (VP securities) and reimburse taxes (tax authorities). They argue that their study provides practical evidence for the feasibility of using blockchain technology in the public sector and is extended to other areas such as the European Value-Added Tax (VAT) system

¹ <https://lympo.io/>.

or disintermediated donations from nongovernmental organizations (NGOs) [109].

Blockchain technology reduces information asymmetry by offering transparent and immutable information to all market participants. This leads to cost reductions and efficiency gains in the marketplace. Furthermore, blockchain has the potential to improve government operations, identity and data management, cybersecurity, and citizen interactions. The impact of blockchain in the public sector extends beyond just improving record keeping and information exchanges and even replaces many of the current administrative roles of governments [110].

In the public sector, a blockchain technology (BCT)-based infrastructure is not intended to replace the current infrastructure. Rather, it is designed to be built by expanding upon existing systems and functional components. Currently, many applications based on BCT beyond cryptocurrency systems combine blockchains and traditional databases (off-chain) [72]. In the public sector, permissioned blockchains appear to be a viable solution to the centralization paradox, as they offer effective enforcement while still allowing governance rules to be controlled by a single organization or consortium. Blockchain-based notarization services improve the auditability of data and increase transparency in administrative processes. The immutability of ledger records may enhance trust in governmental record-keeping among citizens and companies and increase the reliability of markets in which government institutions are involved. In addition to trust and reliability, blockchain provides monetary efficiency gains, such as the streamlining of mortgage handling and land title transfer. Still, public permissionless blockchains have limited use for governments due to the economic and technical limitations associated with them, such as the use of native cryptocurrencies (i.e., a blockchain's inherent token, representing some value such as BTC or Ether), network latency issues, and the potential presence of untrusted participants. Blockchain implementations in the public sector focus on improving efficiency in record keeping and financial management while overcoming bottlenecks in the administrative process [106].

Electronic participation solutions (e-participation) serve a dual purpose of reducing administrative burdens and fostering trust in government processes and record keeping. This is achieved through the use of algorithms and smart contracts that are no longer exclusively controlled by the government [99]. Applications that use smart contracts and blockchain technologies help public managers address and resolve administrative demands, if properly configured and legalized, thus reducing bureaucracy in public services and improving the security, integrity, transparency, and accessibility of information [111].

The blockchain distributes digital information over computers and server networks, updating the information to ensure standardization across all links. This mechanism guarantees transparency and prevents corruption by any single user. The potential applications of blockchain range from online finance and e-commerce to contracting and title registration, accounting, supply chain management, cyber-security, anti-money laundering, file storage, data management, intellectual property rights protection, identity management (e.g., credentials, licenses, etc.), personal records (health, insurance, financial, etc.), asset tracking, contract and vendor management, energy utilities, voting, mitigating and identifying fraud, and the Internet of Things in both business and government sectors [110,112]. Blockchain technology possesses three potential capabilities that transform traditional governance systems, mitigate principal-agent problems—This occurs when two parties such as government (agent) and citizens (principal) agree that the agent will work for the principal in exchange for some incentives [113], and reduce the incidence of document falsification. However, blockchain is believed to have the capacity to improve government systems through increased openness, accountability, and security [114].

The concept of a smart contract was first described in 1997 by the cryptographer Nick Szabo as a type of digital vending machine [115, 116]. When creating a smart contract, developers must ensure that it is error-free, as any errors cannot be corrected once the smart contract is stored on the blockchain. Instead, the flawed smart contract is re-

placed by a new one [72]. Smart contracts formalize governance rules and automatically execute them, resulting in reduced administrative and storage costs, elimination of paperwork, and fewer human errors. Smart contracts also increase the efficiency of allocating public funds to beneficiaries [106]. Using a 'Blockchain platform as a service', smart contracts improve workflows and decision-making processes through artificial intelligence. This automation leads to significant improvements in cost-effectiveness and service quality [117]. Smart contracts require information from the blockchain, or outside of the blockchain, to run [103].

Regular contracts that cannot be read by machines are usually just symbolic and create issues such as ambiguity, making them hard to enforce and resulting in conflicts. Machine-readable smart contracts offer a solution by requiring more detailed specifications from both parties, reducing conflict during the enactment stage [118]. Smart contracts do not require trusted third parties, such as a central authority, entity, or organization, to mediate between parties, and this reduces transaction costs and allows for more automation of transactions [119]. On the other hand, traditional contracts are dependent on third parties and require a lot of paperwork, resulting in high transaction costs [104]. Smart contracts are executed in a distributed way [120], and have various use cases, including lotteries, voting, crowdsourcing, asset sharing, tracking, real estate, insurance, identity management, bidding, rating, gaming, and gambling [116]. For example, banks use smart contracts to issue loans and automate payments, insurance companies use them to process claims, and postal companies that use smart contracts accept payments on delivery [115]. Smart contracts are also used to build DApps, such as decentralized cloud storage (Storj), forecasting markets (Augur), DAOs, or decentralized autonomous corporations (DACs) [121]. Smart contracts fall into three main categories, being static, dynamic, or oracle-driven. Static smart contracts perform basic math operations, such as addition, subtraction, multiplication, and division. Dynamic smart contracts carry out different actions through embedded rules, such as monitoring certain conditions and triggering intended actions. Finally, oracle smart contracts utilize data sources outside the blockchain and are dynamic, including AI oracles. Static and dynamic smart contracts manage data stored on a blockchain [122].

2.5.2. DAO

A DAO, also known as a DAC, is a form of organizational structure that acquires and maintains its operations and executive authority through defined rules, which are computer programs known as smart contracts [123]. DAOs are digital organizations that operate autonomously [124] and are characterized by their transparency and reliance on automated programming codes that are operated by members of the organization. To maintain these attributes, DAOs require an infrastructure that facilitates automation, shared control, validation of decisions and actions, and participation. An appropriate infrastructure for this purpose has been identified as the combination of two emerging technologies, such as DLT/blockchain and smart contracts [125]. A DAO is a blockchain-based system that allows individuals to self-govern and coordinate through a set of self-executing rules deployed on a public blockchain. Popular DAO platforms, such as DAOstack² and Aragon³, define a DAO as a network of stakeholders without a central governing body, regulated by automatically enforceable rules on a public blockchain. Other platforms, such as Colony⁴ and DAOhaus⁵, use different terms such as "colonies" and "magic internet communities" as a proxy for DAOs. The source code of a DAO is distributed on a blockchain with smart contract capabilities, such as Ethereum⁶, typically on a public blockchain [126].

² <https://daostack.io/>.

³ <https://aragon.org/>.

⁴ <https://colony.io/>.

⁵ <https://daohaus.club/>.

⁶ <https://ethereum.org/en/>.

DAOs use blockchain technology to enable transparent and trustworthy community-based decision-making processes within organizations. The leading DAO example is Ethereum, which is built on blockchain technology and has facilitated numerous applications, including the introduction and development of Ether, currently the second largest cryptocurrency after Bitcoin. DAOs improve business strategies by promoting decentralization and empowering actors within an organization. Traditional organizations may view DAO approaches as dangerous or uncontrollable due to lack of oversight and control, which presents a significant risk to stakeholders, revenue, and assets [127]. DAOs operate globally and locally and quickly adapt to changes due to their low collaboration barriers, motivated members, and social awareness [128].

The DAO, built on Ethereum, was the first experiment with decentralized governance that utilized algorithmic authority through cryptocurrency and blockchain technologies. The DAO was aimed at creating a unique social and economic structure. Each DAO should align its structure with the internal goals that are being set. DAOs result in significant savings on legal costs due to their transparency and automatic execution of operations. DAOs are seen as decentralized corporate ledgers for financial and technical cooperation, run autonomously using blockchain technology, with management decisions made using smart contracts [129]. DAOs may pose a challenge to the current conventional role of firms and governments as providers of public and private goods due to the elimination of intricate and risky human control from contracts [72].

Hunhevciz et al. (2022) reveal two main types of DAO: protocol-level and application-level. Protocol-level DAOs are permissionless blockchains that use code to govern stakeholder coordination. On the other hand, application-level DAOs operate on a blockchain that encodes governance rules with smart contracts. The initial DAO application, 'the DAO', was launched on Ethereum but faced catastrophic failure after experiencing a significant security breach that resulted in the theft of millions of dollars. To prevent similar incidents from occurring, new application-level DAOs have emerged, often utilizing frameworks such as Aragon or The DAO Stack that provide reviewed code building blocks that are assembled to minimize the risk of failure as observed in 'The DAO' [125].

Creating a DAO involves human decision-making to automate processes, validate needs, identify opportunities, and find collaborators. A DAO may require human action for off-chain operations, such as management, arbitration, communication, governance, and contracting, which must be responsible. The primary difference between a DAO and traditional organizations is the use of immutable computer program rules that are resistant to attacks. Relations between participants in a DAO are coded and executed digitally and registered in a ledger for decision-making without manual intervention. The management of a DAO is decentralized without top executive managers, relying on automated rules in smart contracts, and its governance functions autonomously through on-chain and off-chain mechanisms that support community decision-making. Following the first failure of the DAO, numerous DAOs have emerged, such as DAOstack⁷, DAOhaus⁸, Moloch DAO⁹, Uniswap¹⁰, BitDAO¹¹, Mango DAO¹², Compound¹³, Radicle¹⁴, Maker DAO¹⁵, Decentraland¹⁶, and Aragon¹⁷. Each DAO has a distinct objective, such as DAOstack proposes a venture capital based on

the Ethereum blockchain, or Democracy Earth¹⁸ works for borderless democracy, promoting public conversation regarding the Internet as a planetary jurisdiction. Decentraland is a metaverse platform that enables contributors to create their own environments and communities recorded on the Ethereum blockchain [130].

Meirobie et al. [114] proposed a blockchain-based framework (Go-chain) that focuses on the government sector and uses DAO and smart contracts to ensure system execution speed. This resulted in a modern and secure electronic document framework for government services, which ensures transparency and increases trust in public services. The framework employs governance rules and tokens defined in the application layer code to operate DAOs. The framework enables the government to securely store all document data without any point of failure and prevents third-party interference to derive profits from the stored data. Smart contracts are used to enable decentralized management of a DAO by implementing transactions, currency flows, rules, and rights within the organization. Members of the DAOs propose, discuss, and vote on organization management through transparent mechanisms. Furthermore, members interact with each other using smart contracts and send or receive tokens [120].

Within a DAO, the project conditions are collectively determined through online proposals and voting, driven by a shared vision. Upon completion of the project, participants are rewarded, fostering an ongoing cycle of action. The participants review the project progress and individual contributions, resulting in the effective motivation of the participants. The DAO uses the decentralization, traceability and immutability features of blockchain technology to establish a trustworthy foundation [131]. Rikken et al. [132] introduced Decentralized Autonomous Citizen Participation Organizations (DACPOs) to facilitate participative budgeting. Participatory budgeting, which involves community voting on budget allocation, is frequently challenged by concerns regarding trust and transparency, which is resolved by using DAO applications that create transparency, execute autonomously, and enable direct citizen control over the budget. DACPOs are used to record and execute data and actions in a decentralized way, increasing transparency, trust, and reducing risks of fraud and corruption. Smart contract-based voting ensures fair and autonomous decision execution without the need for government involvement or centralized bureaucracy. DACPOs offer joint control over storage, transactions, and voting. The results of voting are monitored and executed autonomously, allowing participants to have direct control over the budget without involving government or any third parties. Therefore, DACPO applications are considered suitable for improving citizen participation and trust.

The American CryptoFed DAO1 has been officially registered in the state of Wyoming, marking the first DAO in the region. The concept of DAO has gained significant importance in the development of decentralized software systems based on blockchain technology. DAOs are considered a promising alternative for organizations seeking to adopt decentralized, trustworthy, and transparent governance systems. As a result, DAOs have attracted significant attention from the academic research community. In the quest to improve sharing, security, transparency, and auditability, DAOs are being studied as a viable approach to support organizations, thus allowing their governance or business models to be truly global without any central controlling authority or middleman [133].

2.5.3. Web 3.0

The introduction of ICT and Web 3.0 contributes to the government sector by changing the way government agencies provide advanced and innovative services for the interaction with citizens [122]. The term Web 3.0 was coined by Gavin Wood, the co-founder of the Ethereum blockchain [134,135]. Wood characterized Web 3.0 as providing Inter-

⁷ <https://daostack.io/>.

⁸ <https://daohaus.club/>.

⁹ <https://molochdao.com/>.

¹⁰ <https://uniswap.org/>.

¹¹ <https://www.bitdao.io/>.

¹² <https://dao.mango.markets/dao/MNGO>.

¹³ <https://compound.finance/>.

¹⁴ <https://radicle.xyz/>.

¹⁵ <https://makerdao.com/en/>.

¹⁶ <https://decentraland.org/>.

¹⁷ <https://aragon.org/>.

¹⁸ <https://democracy.earth/>.

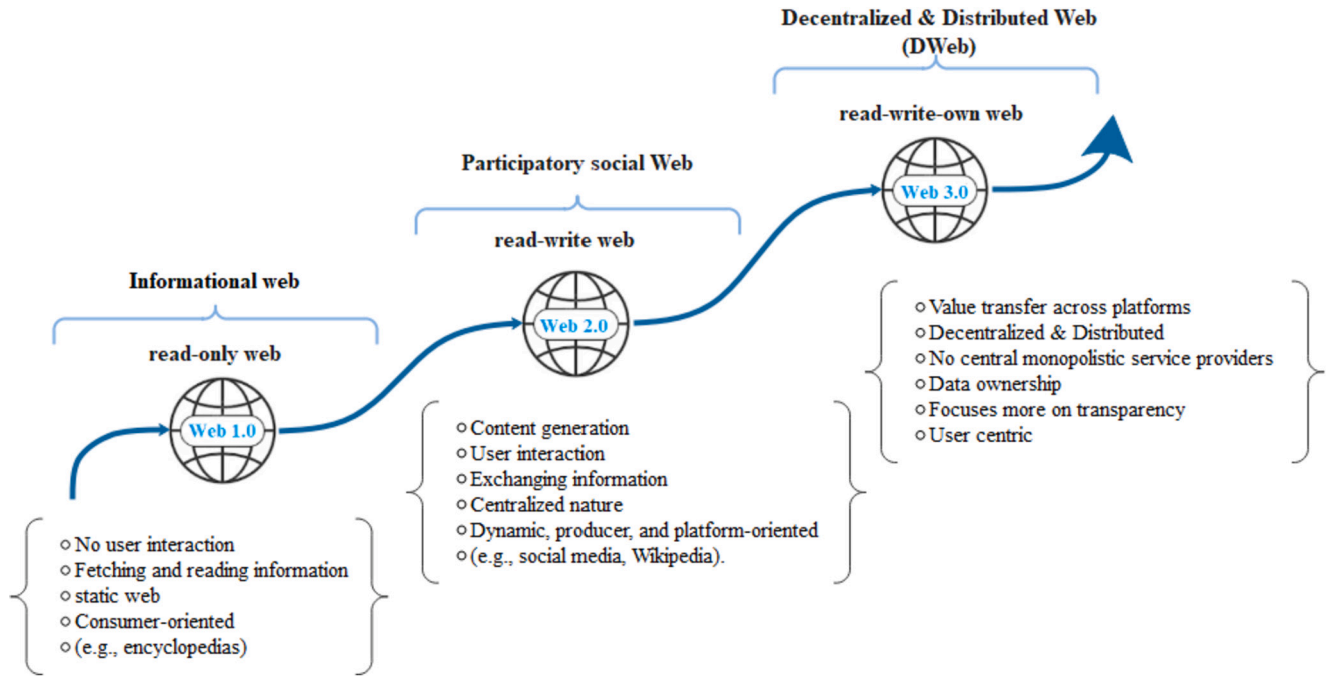


Fig. 1. From static to decentralized web: The Journey of the Internet from Web 1.0 to Web 3.0 (Source: Authors' own creation).

net services to users without the mediation of trusted third parties and giving users full control over their data. A variety of technologies facilitated the development of Web 3.0, while the back-end is supported by the decentralized nature of the blockchain [5].

Web 1.0 (read-only web), where tech companies could publish information through static websites for end users, allowing them to simply read information online without facilitating any interaction (e.g., encyclopedias). Web 2.0 (read-write web), or participatory social web, enables users to produce shared content using server-side scripting to increase online services and allow users to engage more with the platform by providing the possibility of interaction, content generation, and exchanging information (e.g., social media, Wikipedia). The emergence of blockchain marked the third generation of the Internet, Web 3.0 (read-write-own web), which allows the transfer of value, which is decentralized without depending on any central monopolistic service providers. This is called the distributed and decentralized Web (DWeb) [115,136–138], in which the user is the ultimate arbitrator and the main agent for creating and expressing value and is viewed from social, economic, and cultural perspectives [135]. Web 2.0 is vulnerable to attacks and has security issues, namely cross-site request forgery, cross-site scripting, information leakage, and its centralization nature [137,138].

Semantic Web and blockchain technologies serve as two fundamental components of Web 3.0 with the goal of linking data through a decentralized approach [139]. Although Web 3.0 and Web3 are frequently used interchangeably, they denote different concepts. Web3, which is often confused with Web 3.0, is a decentralized version of the Web that leverages DLT or blockchain technologies, prioritizing security and user control by enabling users to own their data and identity. Conversely, Web 3.0 pertains to Tim Berners-Lee's conceptualization of an interconnected or Semantic Web. The Semantic Web, introduced by Tim Berners-Lee as a component of Web 3.0 in 2006, extends the capabilities of the World Wide Web to enable machines to comprehend the meaning and context of information on the Web. It aims to improve efficiency and intelligence by connecting data across various platforms, leveraging artificial intelligence to make the Internet more intelligent. Therefore, Web 3.0 (also known as the Semantic Web or the read-write-execute Web) was introduced by Tim Berners-Lee for a linked Web. On the other hand, Web3 (also known as the decentralized web or the read-

write-own web), introduced by Gavin Wood, co-founder of Ethereum, utilizes blockchain to record public information, protect personal privacy, and eliminate the reliance on trusted third parties through the implementation of smart contracts [140].

Web 3.0 includes technological advancements in areas such as artificial intelligence and the Semantic Web. In contrast, Web3 specifically denotes the decentralized Web that utilizes blockchain technologies to establish "ownable" resources. The combination of Semantic Web technologies (Web 3.0) and blockchain technologies (Web3) has significant potential to enhance the functionality and usefulness of the Web [141]. According to Wood, Web3 is not about cryptocurrencies, blockchain, or token economics, but about decentralization, openness, and transparency [136].

Some of the first Web 3.0 attempts use DAOs, blockchain, cryptocurrencies, and other decentralized technologies to exchange digital assets on the Web. Web 3.0 mitigates issues in Web 2.0, such as over-centralization and dominance by big tech companies, network vulnerability, misinformation, and information disorder; provides more secure, private, scalable, and free information creation, sharing, and transfer; and enables creating, identifying, contracting, and exchanging products (e.g., digital assets). Web 3.0 is enabled by decentralized blockchain technologies, tokens, and DeFi to provide a trustworthy, secure, and immersive decentralized web platform, content, and services. Web 3.0 enhances individual engagement, security, privacy, and scalability and operates in secure, privacy-preserved, and P2P networking, communication, token- and cryptocurrency-based financial services, and ecosystems [142]. A summary of the evolution of the web is shown in Fig. 1.

Web 3.0 is known as the Semantic Web, an extension of the current Web [143], or a modified version of Web 3.0 that is denoted as Web 1.0 Web 2.0 Web 3.0 = Semantic Web [144], in which this Web 3.0 transforms the Web into a readable language that is classified by the system instead of the human [145]. Kuznetsov et al. argued that Web 3.0 is not synonymous with the Semantic Web. Web 3.0 focuses more on transparency and allows one to interact and build trust with multiple stakeholders [146]. Web 3.0 networks are compatible with multiterminal services, providing public access to a custom government system through mobile technologies such as phones, and allowing information to be accessed 24 hours a day in any location [147].

Muthe et al. proposed a Web 3.0-based architecture that removes centralized organizations and supports a completely decentralized, safe, and transparent Internet. The architecture uses IPFS, a P2P distributed hypermedia transfer protocol, to improve reliability, persistence, and speed. A secure decentralized mechanism for initiating data-based payments is created through Ethereum and smart contracts. The proposed architecture employs zero-knowledge proofs and proxy re-encryption to enhance network nodes. Users in a Web 1.0 architecture cannot generate any data and producers have complete control over data access. Web 2.0 architecture narrows the gap between producers and users by offering users limited access to data production. The Web 3.0 architecture aims to ensure that producers and users are indistinguishable in terms of access, sharing, and production rights [148].

Smart contracts are the cornerstone of Web 3.0 operation, and users have absolute control over data privacy. As Web 3.0 allows users to have control over their data, traditional financial institutions are migrating to Web 3.0 due to the rapid development of Web 3.0 payments and its applications, such as DeFi [149].

Web 3.0 is an evolution of digital infrastructure that is supported by blockchain technology. This allows protocol-enforced consensus mechanisms to facilitate the direct exchange of values between users, eliminating the need for trusted third parties. Web 3.0 aims to rebuild the Internet through distributed and decentralized protocols using blockchain technology. This allows for transforming: (1) a redecentralization of the Internet through pushing economic infrastructure into distributed protocols; and (2) digitally building native commercial and economic infrastructure such as money, contracts, organizations, or governance [135].

An example of the transformation of the Web from Web 2.0 to Web 3.0 is Facebook's announcement in June 2019 to establish a new infrastructure to govern its own token, Libra¹⁹ (later renamed Diem), which includes price stability features for its exchange with fiat currencies [83]. With the advent of Web 3.0, everything will be democratized, transforming art, business, and technology; removing intermediaries, and putting individuals more directly in charge of their destiny. The Internet is becoming more participative due to its evolution, and thus facilitates the transition of people from passive consumers to actively engaged producers [150].

The potential influence of Web 3.0 technologies (e.g., blockchain) is unlimited, particularly in the transformation of finance (DeFi), law (data privacy), research (data sharing), new forms of ownership (NFT), and education. Web 3.0 technologies allow immutable and trustless interoperability, enabling powerful new applications to eliminate the need for companies to act as trusted third parties, which has been a major barrier to achieving data security and privacy in the modern era. The Open Science DAO (OPSI) is an example of Web 3.0 technologies for decentralized science working with its community to provide open, discoverable, accessible, interoperable, and repeatable research procedures. Web 3.0 paves the way for a 'trustless' future in which people and devices can communicate to exchange services, data, and value without counterparties that lead to a user-centric, privacy-preserving computing structure for the next generation of the Internet [137].

2.5.4. MFSSIA

The notion of multifactor authentication (MFA) surfaced during the 1980s, yet it encountered limited customer acceptance due to their reluctance to carry additional authentication factors. However, with the advent of smartphones, which enable the transmission of evidence of possession, inheritance, knowledge, or context to another user, MFA has become accessible to billions of individuals, leading to its widespread adoption [151].

The development of government-independent and secure identification and authentication protocols poses challenges and risks.

Table 2

Search index based on research questions.

Primary search keywords	"Blockchain", "blockchain-based", "public ledger", "e-participation", "model", "governance model", "token", "token economy", "smart contract", "DAO", "Web 3.0", "MFSSIA"
Secondary search keywords	"public sector", "public organization", "corruption", "transparency"
Search Query	("Blockchain*" OR "public ledger*" OR "distributed ledger") OR ("e-participation" OR "blockchain model" OR "blockchain-based model" OR "blockchain-based architecture" OR "blockchain-based framework" OR "governance model") AND ("Web 3.0" OR "Web3") OR ("tokenomics" OR "token econom*") OR ("Decentralized autonomous organization" OR 'decentralized autonomous corporation') AND (public sector' OR 'public administration' OR 'public organ *') AND (corruption' OR 'transparency').

Government-controlled systems result in limitations for citizens. Authcoin addresses risks such as data record manipulation and distributed denial of service (DDoS) attacks through its challenge response-based validation and authentication procedure and the blockchain storage system [152]. MFSSIA plays a crucial role in establishing trust in the emerging M2X economy. Existing self-sovereign IA systems lack flexibility in configuring challenge sets. Norta et al. propose a blockchain-based implementation that enables secure data exchange among various smart contract blockchain systems. The MFSSIA system, consisting of decentralized knowledge graphs (DKG) and oracles, facilitates the management of the challenge/response life cycle and the creation of personalized challenge sets [7]. Although the MFSSIA system may have latency issues for fast processing applications, it is beneficial in scenarios such as cross-border trade, reducing risk, time, and expenses [153]. Consequently, using blockchain technology and flexible challenge sets, MFSSIA enables secure engagement between individuals, organizations, and systems in e-commerce transactions.

3. Methodology

To answer the research questions in Section 1, we use the SLR methodology following Kitchenham's guidelines for conducting SLR in software engineering [36], and the systematic approach provided by Dwivedi et al. [154]. Using SLR, we can identify and categorize the latest scientific findings and literature relevant to the current study. The SLR consists of five steps as follows:

- Study Identification
- Study Selection
- Quality Assessment
- Data Extraction
- Study Synthesis

3.1. Study identification

After initial understanding of the corresponding study through reading literature, we identify keywords from research questions, and then these keywords are combined to produce the research string (see Table 2). We use academic databases and websites to identify relevant studies.

3.1.1. Search keywords:

We generate search keywords with the appropriate synonyms from research questions to find the relevant literature. We use 'Blockchain' and its synonyms and e-participation along the governance model as primary search keywords to identify relevant studies based on research questions. Then, we produce secondary search keywords that are more

¹⁹ <https://libra.org/>.

Table 3
Study sources and retrieved paper count.

Search engine	Database	Paper count
Scopus	ACM Digital Library, IEEE Xplore, ScienceDirect, Springer, Taylor & Francis, and Wiley Online Library	65
Google Scholar	ACM Digital Library, IEEE Xplore, ScienceDirect, Springer, Taylor & Francis, and Wiley Online Library	203
Total		268
Total (after duplicate removal)		242

Table 4
Inclusion criteria.

ID	Inclusion criteria
IC1	The article should discuss blockchain, smart contracts, and their role in reducing corruption, improving transparency, and detecting fraud in the public sector.
IC2	The article should focus on blockchain-based model/architecture, governance model, e-participation, smart contract, token economy, Web 3.0, DAO, and MFSSIA.
IC3	The source of the article should be published between 2012-2023.
IC4	The language of the presented information should be English.

specific to the main research questions. Secondary keywords are ‘public sector’, ‘public administration’, ‘corruption’, and ‘transparency’. Next, we produce the search string using the Boolean operators ‘AND’ and ‘OR’ from the primary and secondary search keywords to run the search query.

3.1.2. Study sources:

We search several academic search engines such as Google Scholar and Scopus to obtain relevant literature to form the basis and answer the research questions posed in this study. We identify the necessary journal, conference, and background papers in relation to the blockchain concept, e-participation, governance model, blockchain-based model, corruption, and transparency.

3.1.3. Search process:

We discover the relevant literature from various sources of study. To remove redundant articles, we use Mendeley reference management software. Using its Check for duplicates option in the Tool’s menu of Mendeley Desktop, we search for redundant articles and 26 duplicate articles found in all documents and then removed them manually. Thus, we identify 242 papers published between 2012 and 2023 retrieved from the Google Scholar and Scopus scholarly databases (see Table 3 for details).

3.2. Study selection

To determine relevant studies, we follow a two-phase study selection process that includes inclusion and exclusion criteria (see Table 4 and Table 5). In the inclusion-exclusion phase, we try to identify relevant studies that focus on blockchain-based model and smart contract, and their application in the public sector for the purpose of reducing corruption and achieving transparency. Thus, we distinguish relevant and non-relevant articles in this way and remove unnecessary studies to reduce the number of identified articles. To achieve this, we examine the titles and abstracts of the identified articles and then confirm whether each of the identified articles satisfies the inclusion and exclusion criteria. Of these 242 papers, only 154 have been included for review, and the rest are excluded because they do not meet the predefined inclusion criteria.

Table 5
Exclusion criteria.

ID	Exclusion criteria
IC1	The article focuses on cryptocurrencies such as bitcoin, Ether, etc.
IC2	The article that does not discuss token economy, Web 3.0, blockchain technologies, smart-contract, DAO, and MFSSIA.
IC3	The article that does not discuss blockchain-based models and blockchain application in the public sector.
IC4	The article presents information other than English.

3.3. Quality assessment

We classify all studies and score each article studied according to its relevance. We classify and rate articles on the basis of three-point likert scales using Microsoft Excel and define these three points as follows: 1 (Somewhat relevant), 2 (Quite relevant), and 3 (Highly relevant). Articles that are important for answering research questions are scored 3, articles that provide sufficient information on the research topic and the state of the art are scored 2, and finally articles that contain some basic definitions of the terminologies used in this study are graded as 1.

3.4. Data extraction

In this step of the SLR, we extract the necessary data using the Microsoft Excel tool from primary and supporting studies according to the quality assessment. Two types of data are extracted from studies: (1) data related to publication quality and (2) data required to answer research questions (that is, RQ1, RQ2, RQ3, and RQ4). Details of the data extracted for the primary and supporting studies are presented in Table 6. Specific data on RQs are detailed in Section 4.

4. Study synthesis

This section synthesizes the knowledge from the selected and supporting studies and presents the results derived from the SLR approach, thereby providing answers to the research questions. Section 4.1 provides data to support the answer of RQ1 in Table 7 which includes a summary of existing blockchain-based governance models. Section 4.2 comprises the data in Tables 8 and 9 to support the answer to RQ2 by providing the main components and characteristics of the existing governance models. The answer to RQ3 is provided in Section 4.3, followed by the answer to RQ4 in Section 4.4.

4.1. RQ1: what are the existing blockchain-based governance models?

This study using the SLR approach reveals several blockchain-based governance models that exist in the literature. Table 7 summarizes these models and provides seven models obtained from selected and supporting studies and presents blockchain-based governance models with publication year, digital government interactions, type of sector, context, application, and standard technologies.

Based on the table provided, there exist several blockchain-based governance models in different sectors that serve as G2C or G2B digital government interactions. Of these governance models, the context is general for four of these governance models, while three of them are country-wise, such as India, Pakistan, and Oman. These models are applied in the supply chain, procurement, document verification, and electronic voting with the goal of improving governance, transparency, and accountability in the public sector by using blockchain technology. For example, the “Neural blockchain technology for a new anticorruption token” model is designed for the G2B public sector and focuses on improving procurement processes. The ‘Blockchain-Based Identity

Table 6

List of primary and supporting studies.

No.	Publication title	Type	Year	Ref.
1.	Blockchain Based Identity Verification Model	Conference	2019	[37]
2.	Blockchain technology in supply chain and logistics	Dissertation	2018	[41]
3.	Neural blockchain technology for a new anticorruption token: towards a novel governance model	Journal	2023	[44]
4.	An Auditable and Secure Model for Permissioned Blockchain	Conference	2019	[56]
5.	A novel framework for policy based on-chain governance of blockchain networks	Journal	2021	[57]
6.	Distributed Ledger Technology Based Immutable Authentication Credential System (D-IACS)	Conference	2021	[58]
7.	A conceptual model for e-participation by Omani citizens using blockchain technology	Journal	2021	[59]
8.	Secure Blockchain Interworking Using Extended Smart Contract	Journal	2022	[67]
9.	Implications of the Token Economy: A Taxonomy and Research Agenda	Conference	2022	[68]
10.	Token Economy	Journal	2021	[91]
11.	Token Economy: A New Form Economy with Decentralized Mutual Trust and Collective Governance	Conference	2019	[17]
12.	A Dual Incentive Value-Based Paradigm for Improving the Business Market Profitability in Blockchain Token Economy	Journal	2022	[70]
13.	Stimulative Coordination Models for Cooperative and Competitive Enterprise Alliances Based on Token Economy	Journal	2022	[16]
14.	A Blockchain Token Economy Model for Financing a Decentralized Electric Vehicle Charging Platform	Conference	2020	[85]
15.	Assessment of Blockchain Technology Adoption Factors and Scenarios Within the Economy of Latvia	Conference	2021	[99]
16.	How to design the token reinforcement based on token economy for blockchain model	Journal	2020	[71]
17.	Blockchain-based smart contracts: A systematic mapping study	Conference	2017	[103]
18.	Increasing citizen participation in e-participatory budgeting processes	Journal	2021	[35]
19.	Blockchain and the Public Sector Theories, Reforms, and Case Studies	Book	2021	[105]
20.	Blockchain Applications in the Public Sector: Investigating Seven Real-Life Blockchain Deployments and Their Benefits	Conference	2021	[106]
21.	Blockchain and Trust in a Smart City	Journal	2019	[90]
22.	Blockchain and Regional Workforce Development: Identifying Opportunities and Training Needs	Conference	2021	[110]
23.	A Framework of Blockchain Technology for Public Management in Brazil	Conference	2021	[111]
24.	Blockchain Oracles: State-of-the-Art and Research Directions	Journal	2022	[107]
25.	What about this blockchain thing?	Journal	2022	[108]
26.	Blockchains Unchained: Blockchain Technology and its Use in the Public Sector	Working paper	2018	[112]
27.	Blockchain Enabled Digital Government and Public Sector Services: A Survey	Conference	2021	[117]
28.	The Impact of Blockchain Technology on Finance: A Catalyst for Change	Book	2018	[119]
29.	Blockchain and Web 3.0: Social, Economic, and Technological Challenges	Book	2019	[115]
30.	The Promise of Blockchain for the Construction Industry: A Governance Lens	Conference	2022	[125]
31.	Decentralized trust: Using blockchain technology to facilitate an organizational shift towards autonomous decentralization	Journal	2022	[127]
32.	Decentralized autonomous organizations as the new form of economic cooperation in digital world	Journal	2021	[129]
33.	Governing Decentralized Complex Queries Through a DAO	Conference	2021	[120]
34.	Blockchain Technology as Information Infrastructure in the Public Sector	Conference	2021	[72]
35.	Decentralized identity management: Prerequisite of web3 identity model	TechRxiv Preprint	2022	[5]
36.	Creating Trust in Citizen Participation through Decentralized Autonomous Citizen Participation Organizations (DACPOs)	Conference	2022	[132]
37.	Decentralization and web3 technologies	TechRxiv Preprint	2022	[136]
38.	Web3 & DAOs: an overview of the development and possibilities for the implementation in research and education	Conference	2022	[137]
39.	Blockchain Technology, Cryptocurrency: Entropy-Based Perspective	Journal	2022	[138]
40.	Web 3.0: The Future Architecture of the Internet?	SSRN Preprint	2022	[135]
41.	Decentralized AI: Edge Intelligence and Smart Blockchain, Metaverse, Web3, and DeSci	Journal	2022	[142]
42.	Internet Application and Technology for E-Government Public Services	Journal	2020	[147]
43.	A Blockchain-based, Semantically-enriched Software Framework for Trustworthy Decentralized Applications	Other	2022	[153]
44.	Challenges of blockchain technology adoption for e-government: a systematic literature review	Conference	2018	[52]
45.	A Blockchain Implementation for Configurable Multi-Factor Challenge-Set Self-Sovereign Identity Authentication	Conference	2022	[7]

Verification Model' is targeted at the public sector of G2C in India and provides a more secure and efficient way of verifying documents. The DLT-Based Immutable Authentication Credential System is for the G2C Education sector in Pakistan and is designed to improve document verification for educational institutions. Finally, 'A Conceptual Model for E-Participation by Omani Citizens Using Blockchain Technology' is targeted at the G2C public sector in Oman and aims to improve citizen participation through e-voting. In general, these blockchain-based governance models demonstrate the potential of blockchain technology to improve governance in the public sector and provide a framework for developing best-practice governance models in different contexts.

4.2. RQ2: what are the main components and characteristics of these governance models?

We examine the basic components of existing blockchain-based governance models. These components help practitioners develop efficient and innovative anti-corruption models in the future to involve all stakeholders and avoid possible corruption issues. Furthermore, identifying these blockchain-based governance models enables governments to adopt blockchain-based solutions and avoid possible corrupt cases by eliminating physical interaction with government officials. Physical interaction is avoided by automating business processes, thereby saving cost, time, and reduced manpower. Table 8 presents the com-

Table 7

Summary of existing blockchain-based governance models.

No.	Publication	Year	Ownership type	Context	Application	Std technologies
1.	An Auditable and Secure Model for Permissioned Blockchain [56]	2019	NA	General	Auditing	JVM, akka toolkit, SSL/TLS, DES
2.	A novel framework for policy based on-chain governance of blockchain networks [57]	2021	NA	General	Decision-Making	PBM, REST API, JSON, asymmetric key pair, DID, VM
3.	Neural blockchain technology for a new anti-corruption token: towards a novel governance model [44]	2022	public sector	General	Procurement	Cryptographic keys
4.	Blockchain technology in Supply Chain and Logistics [41]	2018	NA	General	Supply chain	ERP, TMS, SCNs
5.	Blockchain Based Identity Verification Model [37]	2019	Public Sector	India	Document verification	GO, asymmetric cryptography
6.	DLT-Based Immutable Authentication Credential System (D-IACS) [58]	2021	Education Sector	Pakistan	Document verification	GO, Java
7.	A Conceptual Model for E-Participation by Omani Citizens Using Blockchain Technology [59]	2021	public sector	Oman	e-voting	NA

Table 8

Summary of components found in the existing governance models.

No.	Component	Description	Provide
1.	Dual-Key Pair Mechanism Distributed Key Generation Protocol and Threshold Paillier Cryptosystem [56]	Dual-key pair satisfy encryption, integrity, authentication, and nonrepudiation simultaneously. DKG needs multiple parties to generate keys using secure multiparty computation.	Threshold Paillier cryptosystem enables key management decentralization and ensures confidentiality.
2.	Protocol layer, Protocol_Module Virtual Machine, Voting Contract, Proposal Contract, Decentralized-Identifier(DID) Registry-Contract, X Contract [57]	The Protocol_Module serves as the main component of a blockchain node, Virtual Machine ensures safe execution of smart contracts. The Voting Contract accepts proposals, and the Proposal Contract manages voting procedures. The DID store actors' identities in governance. Governance policies are managed through the Policy Enforcement Point (PEP) interface using the X Contract.	A means of streamlining Parallel and Distributed Computing Architecture (PDCA) management while also enabling automated translation of policies into commands and configuration parameters that are comprehended by PDCA devices.
3.	Blockchain as a Service (BaaS), Smart Contracts, Front-End / Devices, Virtual Token, DocKeeper, NDL EQUO BackEnd, Database [44]	BaaS ensures scalability and resolves distribution issues. Smart contracts automate record management, secure data, and regulate access at each stage of the e-procurement process. The block generates a virtual token that contains information about the administrative process. DocKeeper dApp allows authorized agents to access administrative records as indicated in the smart contract. NDL EQUO BackEnd comprises the framework, server, connectors, and database to store institutional records.	Effective measures to prevent corruption in public procurement. This solution engages stakeholders, resolves security and transparency concerns, and mitigates the risk of corruption in administrative processes.
4.	User Interface-Mobile app for tracing IoT Sensors Smart contracts [41]	The design of the application's user interface plays a vital role in the usability of the blockchain solution. Internet of Things (IoT) sensors automate the process and reduce errors by tracking the conditions of the product. Smart contracts allow network participants to build terms, conditions, and other logic into digital contracts and improve capital circulation in the supply chain.	Transparent agricultural supply chain solution transparent using blockchain technology.
5.	Hyperledger, Interface, File handling/ Database (IFS) [37].	Hyperledger ensures confidentiality, flexibility, and scalability for distributed ledgers. Authorized parties access the blockchain interface to verify the authenticity of the document. IPFS stores files with their hash values being shared on the blockchain.	Streamlined government procedures, full digitalization without physical interaction with officials, cost and time savings, and reduced manpower.
6.	Calling Layer, legacy systems Adaptive Layer, Hyperledger DLT Layer, DLT database Smart contracts [58]	Calling Layer enables interactive systems operating on legacy systems. The DLT database stores the credentials information of the 'Adaptive Layer'. The smart contract encompasses the system's application logic.	Trust, increase efficiency, promote transparency, reduce costs, and eliminate intermediary.
7.	e-Participation actors, Dashboard, user evaluation and feedback, System Interface Middleware, database, governance authority, citizen registration, actors verification, activities [59]	Using Blockchain technology, ballot papers are encrypted with a cryptographic key that can only be decrypted using voters' private keys.	enhanced citizen participation in policymaking and representative institutions.

mon components found in existing blockchain-based models, providing a brief description of each component along with the solutions offered by each.

The components in Table 8 indicate that only three blockchain-based governance models out of seven are proposed with the goal of achieving transparency and combating corruption. First, the gover-

Table 9
Characteristics of existing governance models.

No.	Governance model	Characteristic
1.	Novel multi-supervised permissioned blockchain (MSPB) governance model [56]	Supports auditing and eliminates centralized entities in the system, and avoids information security threats caused by centralized management entities.
2.	A novel on-chain governance model [57]	Blockchain cannot tolerate service interruptions for any reason. Thus, interruptions caused by hard forks or soft forks should be avoided. To this end, PBM has the ability to change the behavior of a blockchain node or even smart contracts without stopping them. PBM fits better with the governance of private or permissioned blockchain networks, which are a class closer to legacy information systems. Policy-based governance deploys updates immediately, after the end of the voting period, without the need for an additional process.
3.	Federated permissioned model [44]	Accelerates the transactions, so they are used in e-government systems by many users and operators. The use of smart contracts to manage data in administrative processes automatically and avoid corrupt practices in the administration through enhanced security and traceability.
4.	Blockchain-based governance model in the supply chain [41]	Mitigates the fear of stakeholders of sharing their sensitive data. The private ledger protects sensitive data through selective permissions. Communication and consensus mechanisms ensure that all participating nodes jointly ensure the safety and authenticity of data stored on the blockchain and collectively maintain the openness and transparency of the data in the blockchain.
5.	Blockchain Based Identity Verification Model [37]	Improves the efficiency of the document verification and issuing process drastically and achieves security, reliability, and transparency.
6.	DLT based model (D-IACS) [58]	Supports and facilitates the conventional transformation of existing systems toward DLT systems based on the fourth industrial revolution (4IR) that accommodate smooth communication between legacy systems and the latest systems.
7.	Blockchain-based e-participation model of Omani citizens [59]	Enhance citizen engagement, reduce costs, and increase the level of trust. Address scalability, minimize system complexity, and increase transparency to generate greater user interest in e-participation.

nance model proposed by Francisco et al. [44] provides efficient and innovative anticorruption solutions in public procurement ensuring the participation of all stakeholders and avoiding possible corruption issues during administrative processes, by addressing security and transparency issues. Second, in the model presented by Malik et al. [37], an efficient government, automated processes, and a completely digital process are achieved, thus eliminating physical interaction with government officials, resulting in a more efficient government, cost-efficiency, time savings, and reduction in manpower. The final model, suggested by Janjua et al. [58], promotes trust and transparency, increases efficiency, reduces costs, and eliminates intermediaries. Therefore, using the mentioned models, governments streamline public procurement processes, enhance transparency and accountability, and increase public trust in their institutions, ultimately resulting in better outcomes for all stakeholders involved.

From the characteristics of the existing governance models described in Table 9, it is evident that blockchain-based solutions offer a wide range of benefits, including increased security, transparency, efficiency and trust.

4.3. RQ3: what is a blockchain-based token economy and its role in eliminating trusted third parties in transferring ownership?

Research demonstrates that a blockchain-based token economy is a decentralized system that leverages the tokenization process to convert the value of an asset into a digital form. This digital representation of assets, including physical or electronic ones, is facilitated over the blockchain, thereby eliminating the need for intermediaries such as banks, notaries, and lawyers, which enhances transaction efficiency and cost-effectiveness.

The study demonstrates that in this system, tokens, which are created and managed through a DLT protocol or smart contracts, serve as a unit of value. These tokens represent various forms of value and rights, including ownership, and effectively map real economic activity to digital economic activity. The research also underscored the advantages of the decentralized nature of a token economy, which allows multiple actors to participate in decision-making processes related to asset ownership management without a central authority. This approach offers superior liquidity, faster and less expensive transactions, greater transparency, and greater accessibility. It also strengthens public trust in

Table 10
Comparison between traditional system and token economy.

Aspect	Traditional system	Token economy
Authority	Centralized	Decentralized
Transaction speed	Slow	Fast
Transaction cost	High	Low
SPOF	Present	Minimized
Asset ownership	Managed by third party	Managed by DLTs
Security	Lower	Higher
Efficiency	Lower	Higher
Flexibility	Lower	Higher
Transparency	Low	High
Accessibility	Limited	High
Risk of fraud	High	Reduced
Public trust	Variable	Strengthened

organizations with low degrees of trust by ensuring openness, tamper-resistance, and transaction traceability.

Furthermore, the research indicates that a blockchain-based token economy addresses the disadvantages of traditional asset ownership transfer methods, such as high costs, slower processing times, and the existence of an SPOF (see Table 10). By allowing agents to digitally represent and manage ownership of assets using tokens on a decentralized platform, transactions become faster, cheaper, and less susceptible to fraud. Token economies offer organizations new ways to raise funds and empower stakeholders to democratize business models, and automate financial functions such as lending, borrowing, and insurance through tokens and smart contracts without the need for central intermediaries. They also play an important role in the public sector, mostly in the form of utility tokens.

The study concludes that a blockchain-based token economy has the potential to revolutionize the way we exchange value and transfer ownership of assets. By eliminating the need for trusted third parties, token economies accelerate transactions, reduce costs, and enhance accessibility. However, the research also emphasized that to successfully establish a token economy, technological, standardization, and governance aspects must be taken into account.

Table 11
Impact of blockchain and its subset technologies on public service.

Technology	Effect
Blockchain	Enhances transparency and minimizes fraud and errors in public service delivery. Improves service delivery efficiency, transparency, and simplifies bureaucratic processes.
Smart contract	Reduce administrative costs, eliminate paperwork, automate governance rules, and streamlines the allocation of public funds to beneficiaries.
DAO	Enable transparent and trustworthy decision-making within organizations. Facilitates decentralized decision-making, fosters community-based governance, and empowers direct citizen control over budget allocation in participatory budgeting.
Web 3.0	Facilitates the direct exchange of values between users, ensures user control over data, and eliminates third-party dependence. Contributes to the government sector by transforming service delivery. Mitigate vulnerabilities and security issues associated with Web 2.0.
MFSSIA	Establish trust in the M2X economy. Facilitates secure data exchange. Addresses risks related to data manipulation and DDoS attacks.

4.4. RQ4: to what extent does the use of smart contract, blockchain technologies, DAOs, web 3.0, and MFSSIA affect the public sectors to achieve transparency?

By automating and transparently executing rules and governance on a blockchain platform, smart contracts eliminate the need for intermediaries, reducing the possibility of corruption or manipulation by third parties. Additionally, smart contracts offer greater traceability and auditability, since all transactions and decisions made are recorded on the blockchain and are immutable. In addition, smart contracts reduce administrative and storage costs, eliminate paperwork, and prevent human error, leading to more efficient and cost-effective allocation of public funds to beneficiaries. The use of artificial intelligence in conjunction with smart contracts further improves decision-making processes, resulting in higher service quality. Smart contracts reduce conflict and ambiguity in traditional contracts since they require more detailed specifications from both parties and execute those specifications automatically, minimizing the possibility of disagreements arising during the enactment stage.

The decentralization and transparency offered by blockchain are instrumental to eliminating bureaucracy and paperwork and ensuring that all recorded information on the network is free from fraud and errors. Blockchain technology is instrumental in detecting and reducing or eliminating corrupt activities by providing transparency, which is lacking in current centralized systems. Blockchain-based notarization services improve the auditability of data and increase transparency in administrative processes, while the immutability of ledger records may enhance trust in the management of government records among citizens and companies.

The use of blockchains and their subset technologies in the public sector demonstrates significant potential to achieve transparency by ensuring safe and efficient transactions without intermediaries, as shown in Table 11.

The integration of blockchain has shown notable improvements in the delivery of public services, including streamlined land registration processes, enhanced auditability of government operations, and accessible notarization services. Additionally, the implementation of smart contracts has led to automated administrative processes, reduced transaction costs, and increased transparency within various sectors, such as real estate, insurance, and financial services.

DAOs, as a form of organizational structure that uses defined rules, computer programs known as smart contracts, and blockchain technology to function autonomously, eliminate the need for a central governing body, which ensures transparency in decision-making processes.

DAOs may pose a challenge to the conventional role of firms and governments as providers of public and private goods. Nevertheless, several DAOs have emerged with different objectives, such as venture capital, borderless democracy, and metaverse platforms that align with internal goals. DAOs have emerged as key enablers of transparent and decentralized decision-making processes within the public sector. Although DAOs offer benefits such as reduced administrative costs and improved trust and accountability, challenges related to legal compliance, data security, and technological standardization require careful management for their effective implementation. The introduction of Web 3.0, as the next generation of the Internet, has the potential to revolutionize the public sector by providing advanced and innovative services without relying on central authorities. This transformative technology fosters transparency and trust by eliminating the need for third-party intermediaries and by empowering direct interactions between individuals and organizations. Therefore, the emergence of Web 3.0 is transforming the way governments offer services to citizens by introducing advanced and innovative methods of interaction.

The MFSSIA is used in scenarios such as B2B-commerce where the identity authentication process is bureaucratic and time-consuming for individuals, organizations, and systems. MFSSIA has proven to be effective in improving security and trust within the public sector by using multiple factors to authenticate user identities. By addressing risks related to data manipulation and security breaches, MFSSIA ensures the secure participation of individuals and organizations in various transactions, thereby facilitating more efficient operations and improved service delivery. Therefore, the use of MFSSIA affects the public sector to some extent in achieving transparency, but its effectiveness depends on the specific context and requirements of the system.

The utilization of blockchain, DAOs, and token economy in government operations presents several challenges:

- **DAOs and government roles:** DAOs could lead to more efficient and transparent government services, and also introduce the risk of more complex and unpredictable contracts. For example, a DAO-based participatory budgeting system could allow citizens to vote directly on public fund allocation, leading to more transparent and accountable government spending. However, this could also result in more volatile and unpredictable spending patterns due to the absence of human judgment inherent in traditional government budgeting processes.
- **Complying with legal and regulatory frameworks:** DAOs have to adhere to the existing laws and regulations that govern the public sector and address potential conflicts and disputes that may arise from their operations.
- **Ensuring data privacy and security:** DAOs must protect the sensitive and personal data of their members and stakeholders and prevent unauthorized access and malicious attacks on their systems.
- **Achieving technology standardization and interoperability:** DAOs must use compatible and reliable technologies that enable communication and integration with other systems and platforms.
- **Business challenges in token economy:** The business challenges of the token economy, such as low liquidity and lack of clear regulations. The token economy is a system where digital tokens are used as a medium of exchange for goods and services. This can create new opportunities for innovation and value creation. However, it can also pose significant challenges for businesses, such as dealing with volatile markets, complying with unclear or inconsistent regulations, and ensuring the security and legitimacy of tokens.
- **Blockchain adoption challenges:**
 - Technological challenges include security vulnerabilities, scalability issues, lack of system integration and interoperability, and low flexibility.

- Organizational challenges include acceptability issues, organizational complexity and misalignment, lack of governance model, and cybersecurity risks.
- Environmental challenges include lack of legal and regulatory support and high energy consumption.

Overall, the successful implementation of blockchain, smart contracts, DAOs, Web 3.0, and MFSSIA in the public sector requires strategic management to address challenges such as legal and regulatory compliance, data privacy protection, security, and technology standardization. The extent of their impact on enhancing transparency within the public sector relies on how these technologies are developed and implemented and on how the associated challenges are addressed.

5. Discussion

The findings of this study suggest that only three existing blockchain-based governance models are more appropriate to achieve transparency and combat corruption in the public sector. These models are applied in various sectors, such as procurement, document verification, and e-voting. The first model [44] is proposed to prevent corrupt practices in the administration, improve procurement processes, and increase stakeholder participation. This model is more appropriate if it is used in e-government systems by citizens and public servants. The second model [37] is designed for the public sector to provide a more secure and efficient way to verify documents without any physical interaction with public servants using asymmetric cryptography, thus reducing fraud by eliminating the need for physical document verification. Similarly, the third model [58] is designed to improve the verification of educational documents for educational institutions. It uses a distributed ledger system to ensure the immutability of educational credentials. The last model [59] is designed to improve citizen participation in electronic voting through a secure and transparent voting system using blockchain technology, thus minimizing voter fraud and increasing public trust in the electoral process. Our analyses indicate that none of the existing blockchain-based models focus on participatory budgeting using smart contracts, blockchain, DAOs, Web 3.0, and MFSSIA, while these modern technologies play a potential role in addressing corruption and transparency, data, and asset ownership. The components of existing models ensure the security, transparency, and immutability of the data stored on blockchains. The scalability problems of blockchains raise concerns related to large-scale data processing in the public sector.

Research highlights that the token economy is created and managed using a DLT protocol or smart contracts, and that the instances of the token economy are centralized or decentralized [68]. The research highlights the importance of the token economy in the public sector, mostly in the form of utility tokens. The use of tokens and smart contracts in the public sector automates financial functions such as lending, borrowing, and insurance, without the need for central intermediaries.

The findings emphasize the transformative potential of blockchain technology within the public sector, which offers greater transparency, efficiency, and reliability. In particular, the application of blockchain in land registration, government operations, and notarization services demonstrates its ability to streamline bureaucratic processes, minimize errors, and ensure secure and immutable data records. Similarly, the implementation of smart contracts is highlighted as a means of automating administrative tasks, reducing costs, and fostering transparent transactions, thus improving public service delivery. The findings demonstrate that the use of smart contracts, blockchain technologies, DAOs, and Web 3.0 significantly enhances transparency in the public sector. Smart contracts operate on top of blockchain technologies to automate the execution of rules and governance on a blockchain platform, thereby eliminating paperwork and the need for third-party intermediaries, reducing administrative and storage costs, addressing corruption and conflicts in contract execution, increasing traceability and auditability, preventing human errors, and improving decision-making processes. Similarly,

blockchain technology enhances transparency in the public sector by eliminating bureaucracy and paperwork, ensuring that all recorded information on the network is free from fraud and errors, detecting and reducing corrupt activities, and improving government operations, and data management. Blockchain-based notarization services improve the auditability of data and increase transparency in administrative processes, while the immutability of ledger records may enhance trust in government record keeping among citizens and companies. Several studies in Refs. [106,110] emphasize that blockchain-based solutions do not endanger the role of government entities as intermediaries but rather serve as complementary or partial substitutes for current public services.

Furthermore, the study underscores the significant role of DAOs in facilitating decentralized decision-making and improving trust and accountability. DAOs affect the public sectors to achieve transparency by relying on automated programming codes that are operated by members of the DAO organization, eliminating the need for a central governing body such as high-ranking public servants and enabling trustworthy community-based decision-making processes within organizations. DAOs yield significant savings on legal costs due to their transparency and automatic execution of operations. DAOs promote decentralization and empower participants within an organization. Conventional organizations may perceive DAO approaches as dangerous due to the absence of oversight and control, posing a significant risk to stakeholders, financial stability, and valuable assets [127]. The authors of Ref. [72] claimed that the introduction of DAOs potentially disrupts the established roles of companies and governments in the delivery of public and private goods, since they eliminate the need for human control in contracts, which is a complex and risky process. Participatory budgeting, a process where the community votes on budget allocation, often faces trust- and transparency issues. However, these concerns can be addressed by using DAO applications that establish transparency, operate autonomously, and allow citizens to directly control the budget. To address trust and transparency issues, a decentralized autonomous citizen participation organization (DACPO) is designed in Ref. [132] to facilitate participatory budgeting that records and executes data and actions decentralizedly, ensuring fair autonomous decision-making without government involvement. This approach aims to achieve transparency, trust, and minimizes the risks of fraud and corruption.

The decentralization and ownership characteristics of Web 3.0 enable users to have complete control over their data, ensuring security and privacy. Kuznetsov et al. [146] argued that Web 3.0 should not be confused with the Semantic Web as they have distinct characteristics. Web 3.0 places greater emphasis on transparency and facilitates interaction and building trust among various stakeholders. Web 3.0 signifies a crucial advancement in facilitating decentralized exchanges and promoting transparency within the government sector. Its ability to provide innovative services, promote trust, and transform various domains, including finance, law, and education, underscores its potential to revolutionize traditional bureaucratic processes and empower direct interactions between individuals and organizations.

The findings demonstrate that the implementation of MFSSIA significantly improves security and trust within the public sector. Using multiple authentication factors and blockchain technology, MFSSIA addresses potential risks and vulnerabilities, ensuring secure participation in e-market transactions and facilitating more efficient cross-border trade operations.

Consequently, as future research, we consider investigating the role of e-participation in e-participatory budgeting projects with means of smart contracts, blockchain technologies, DAOs, Web 3.0, and MFSSIA to achieve greater transparency, tackle corruption, and increase the level of citizen e-participation in public procurement to reduce the level of corruption. Currently, many e-participation models exist in order to facilitate and empower citizen engagement with the government, and these e-participation models do not provide users with control over their data. Thus, users are discouraged from participating

in policy-making. To this end, future studies should take into account the integration of a blockchain-based e-participation model employing Web 3.0 to empower users (citizens) and increase their freedom, providing full access to their own data through SSI.

6. Conclusion

Blockchains, including their subset technologies, such as smart contracts, Web 3.0, and DAOs, have the potential to enable governments to combat corruption and improve transparency by minimizing physical interactions between citizens and public servants. In our research, we focus on existing blockchain-based models, examining their fundamental components and characteristics. We also delve into the role of token economies in eliminating intermediaries and discuss the effects of blockchains and their subset technologies, along with multifactor SSI authentication, within the public sector. Through an exhaustive systematic review of the literature that includes 242 peer-reviewed journals, proceedings, and book chapters published between 2012 and 2023, we identify seven blockchain-based governance models that emphasize improving governance and transparency in the public sector.

Our findings indicate that three of these blockchain-based models offer anti-corruption solutions in the public sector. By eliminating physical interaction with public servants and eliminating intermediaries from the workflow, these models effectively engage stakeholders and promote transparency. Additionally, our study highlights the potential of token economies to provide organizations with new avenues for fundraising. By eliminating the need for trusted third parties, token economies empower stakeholders to digitally represent and manage the ownership of assets using fraud-resistant tokens, independent of central actors and their platforms. In addition, our findings demonstrate the significant potential of blockchain technologies, along with their subset technologies such as smart contracts, Web 3.0, and DAOs, to combat corruption and improve transparency within the public sector. These technologies achieve these goals by preventing system manipulation, increasing traceability and auditability, eliminating reliance on intermediaries, reducing paperwork, and granting users ownership and control over their data.

This paper is subject to a limitation in terms of its heavy reliance on existing published literature, which introduces the possibility of publication bias. Additionally, the scalability of blockchain-based governance models and the ethical implications associated with token economies have not been extensively explored in this study. To address these limitations, future research should focus on conducting studies that examine real-world implementations of blockchain-based governance models. Furthermore, investigating the impact of these models in specific sectors, such as public procurement and municipalities, particularly in the context of participatory budgeting projects, will contribute to a deeper understanding of their potential benefits and challenges within specific contexts.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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